Chapter 13 worked solutions • I/O systems

OS13-E01 • PIO publication and lost wakeups

1. CPU sets READY; controller observes it, sets BUSY, and reads old DATA; CPU only then writes new DATA. Completion cannot repair the command's old input. Required I/O accessors/barriers must make data visible before command publication.
2. Waiter reads false; IRQ sets true and wakes before any waiter exists; waiter sleeps. Under the same lock protect predicate and wait queue. While !done, enqueue and atomically release the lock while sleeping. IRQ sets done/wakes under corresponding locking. The while loop handles unrelated wakeups. IRQ count is not a completion predicate.

OS13-E02 • Partial vector writes

1. First consume120 then50, leaving[base1+50,30],[base2,300], total330. Next consume30 then100, leaving[base2+100,200]. Delivered300 plus remaining200 equals original500.
2. Retain that progress and wait for writability or follow cancellation rules. Resending duplicates300 B. EAGAIN does not undo earlier successful transfers.

OS13-E03 • Virtual timers

1. Initially program7. Insertion gives[6D,7A,10 B,10C], so reprogram6; canceling A leaves6 unchanged. At6 remove D and program10; at 10 remove B and C. A late handler processes every deadline$\leq$now, not only one equal to now.
2. No. Hardware/IRQ delay, deferred work, and scheduling can delay callback execution. The trigger request is not unconditional user-thread execution at an exact instant.

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OS13-E04 • DMA cancellation and reclamation

1. Hardware may still write dma_addr. Reallocated pages could belong to another process. Only confirmed cessation permits synchronization, unmapping, and release.
2. Under a common lock/CAS, change IN_FLIGHT to DONE or ABORTED; only the winner publishes the terminal result. Two winners would require the second to transition from an already terminal state, contradicting its precondition. CAS proves at-most-once publication; it does not stop DMA. Here normal completion or successful abort acknowledgment guarantees hardware quiescence; perform the required platform synchronization before unmapping. This safety proof does not guarantee a broken device eventually responds.
3. No. Late abort/timer/IRQ callbacks may still inspect state and must already own stable references. With owner/IRQ/abort count3, IRQ exit gives2, owner exit1, abort observing DONE and exiting0; only then free. User timeout notification, DMA release, and control-block reclamation are different events. If DMA directly uses a user buffer, the API must require it to stay valid and inaccessible to the caller until quiescence and synchronization. A kernel-owned bounce buffer can instead decouple the user buffer, subject to the copying and API contract.

OS13-E05 • Finite double-buffer batch

1. Under the specified earliest-start schedule, A 1 inputs0–3/processes3–8; B2 inputs3–6/processes8–13; A 3 inputs8–11/processes13–18; B4 inputs13–16/processes18–23. Input waits for a free slot.
2. One buffer takes 4(3+5)=32 ms. Two take3+5+3 • 5=23 ms. Steady period5 ms is CPU-limited. Four times5=20 omits initial 3 ms filling. Input and processing cannot overlap within one buffer, only across buffers.

OS13-E06 • STREAMS watermarks

1. Acceptance reaches900 and stops upstream. One service yields600, still not below400; another yields300 and back-enables. This model explicitly uses strictly below, rather than another implementation's≤.
2. Immediately after the second service and before upstream resends, downstream 300 B plus upstream 600 B totals 900 B. Backpressure moves waiting upstream; an unpausable source needs dropping or throttling when all queues fill.

OS13-E07 • Resource bounds and Little's relation

1. Device60000/s; CPU=1/(10 • 10^{-6})=100000/s; memory=2 • 2^{30}/(24 • 2^{10})=87381.33/s. Minimum60000/s is the upper bound, limited by device. Bounds need not be attained.
2. L=50000 • 0.0004=20; in service=50000 • 0.0001=5; queue15; wait0.0003 s. Service occupancy is the aggregate across parallel servers, not the occupancy of a single serial server. The same steady workload and boundary apply.
3. No. At least10000 requests/s accumulate, invalidating finite steady state. Fixed L=20 cannot follow from Little's relation when the queue grows without bound. GiB means2^{30} B.

OS13-E08 • Naming, LBA, and durability

1. 12345=3 • 4096+57. PartitionLBA=90 • 8=720; whole-disk start1744. The requested bytes 57–156 lie entirely in whole-disk sector 1744, at inclusive sector offsets 57–156. The entire block spans sectors 1744–1751; a whole-block cache fill would read that range. The application request does not fix the hardware transfer length.
2. No. The API accepted100 B, possibly only in dirty RAM or volatile controller cache. Compatible filesystem synchronization and device flush/FUA contracts establish durability.
3. No. Without persistent queryable device deduplication/progress, the same host ID can print again. Missing acknowledgment cannot distinguish unprinted from printed. Add a device contract or report outcome unknown.

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz, Galvin and Gagne, Operating System Concepts, 9th ed., Chapter 13.

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